Constraining Black Hole Growth in the JWST Era

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Abstract

Aims. We assess which JWST-identified accreting black holes and candidates at $z \geq 4$ place meaningful constraints on early growth histories, and how these constraints depend on seed formation, radiative efficiency, and reported mass uncertainty.

Methods. We compile a provenance-aware catalogue containing 350 literature measurements across 32 source families, represented by 338 object records in 337 host systems pending final identity reconciliation. After excluding all six records associated with three unresolved identity groups, our primary comparison uses 224 objects with comparable mass methods and accepted evidence classifications; the exploratory comparison includes 234 numerical objects. The full catalogue retains all 237 mass-bearing records and 101 records without eligible masses. We propagate published asymmetric mass errors using 10,000 deterministic Monte Carlo draws and preserve mass-method systematics as separate envelopes. We compare fixed seed, formation-redshift, and efficiency scenarios using a reproducible catalogue workflow.

Results. Under the reference assumptions $z_{\text{seed}} = 30$, $\epsilon = 0.1$, no merger boost, and a $10^2\,M_\odot$ seed, 12 primary-sample objects require a lifetime-average $f_{\text{Edd}} > 1$ at their point estimates. Eight remain above unity at the 16th percentile, and six have $P(f_{\text{Edd}} > 1) \geq 0.95$. The corresponding exploratory counts are 14, ten, and eight. UNCOVER-20466 has the highest required average rate, with median $f_{\text{Edd}} = 1.453$ and a 16th–84th percentile interval of 1.354–1.549. Increasing the seed mass to $10^3\,M_\odot$ reduces the primary count above unity to four; delaying seeding to $z_{\text{seed}} = 20$ increases it to 22. At the ideal non-spinning thin-disk efficiency $\epsilon = 0.05719$, no point estimate in either sample requires $f_{\text{Edd}} > 1$ under otherwise unchanged assumptions. The inferred growth requirements are therefore conditional on the adopted model; heterogeneous selection functions and mass estimators preclude interpreting global rankings as demographic measurements.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

The discovery of massive black holes in the early Universe raises a fundamental question: what growth histories could have assembled them within the available cosmic time? The strength of this constraint depends on both the observed black-hole mass and redshift, as well as assumptions about seed formation, radiative efficiency, and the duration of active accretion. An apparently demanding object under one set of assumptions may therefore be compatible with less extreme growth under another.

Here, we use a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates to ask which objects meaningfully constrain early growth histories. The atlas covers $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We calculate the average accretion rates required across a range of seed and growth assumptions, incorporating reported mass uncertainties and distinguishing measurements with different evidential and methodological limitations. Source-native measurements, identity links, and mass provenance are retained to support comparison across heterogeneous literature samples.

Our aim is to identify which growth requirements remain demanding across plausible scenarios, which depend primarily on uncertain assumptions, and where improved observations would most strengthen the constraints. This approach treats the catalogue as a means of testing possible growth histories, rather than assuming that every high-redshift black hole presents the same assembly problem. The constraints apply to individual objects under stated assumptions; we do not estimate a pooled mass function, occurrence rate, or formation-channel fraction.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, Dayal [2] considered light and heavy seeds at $z_{\rm seed} = 25$ and explored a primordial-seeding interpretation. Here we use this established growth framework to assess how object-level constraints change when published masses, measurement methods, and growth assumptions are treated consistently across a larger evidence atlas. The contribution is a traceable comparison of which constraints persist under reported mass uncertainty and which disappear when assumptions change; we do not infer a primordial abundance or introduce a new growth mechanism. Survey studies provide a complementary perspective: the JADES broad-line census [3] measures current accretion and host scaling relations, while the narrow-line census [22] broadens the evidence for obscured activity. Our comparison asks what cumulative growth the adopted masses require, which is distinct from measuring present accretion or counting AGN in a survey. It uses those source measurements without imposing a common selection function. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH} c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \tag{1}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M}_{\rm BH} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon}\frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{2}$$

The inflow rate is defined at the radiating flow; additional mass loss before that point is not modelled. At fixed efficiency, define

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f_{\rm Edd}} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt. \tag{3}$$

The analytic growth relation adopted by Dayal [2] then becomes

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}\right], \qquad t_{\rm efold} = \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{4}$$

The e-folding time assumes constant positive $f_{\rm Edd}$; it is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$. The dimensionless $B_{\rm merge}$ represents an effective multiplicative merger boost, not a resolved merger history.

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{5}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected. In particular, the supplementary $z_{\rm seed} = 3400$ comparison extrapolates this age relation and is not a radiation-era seed-formation calculation.

Two inversions provide the principal growth diagnostics:

$$\overline{f_{\rm Edd}}_{\rm req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right], \tag{6}$$

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{7}$$

In Eq. 7, $M_{\rm seed,req}$ denotes the required seed mass. The zero floor in Eq. 6 follows the implementation: if the boosted seed already exceeds the observed mass, no positive accretion is required, but this does not make that fixed history an exact mass match.

For a two-state history with the same efficiency in both states,

$$\overline{f_{\rm Edd}} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f_{\rm Edd}}_{\rm req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{8}$$

Here D is the active-state time fraction, $f_{\rm burst} > f_{\rm quiet} \geq 0$, and the diagnostic requires $\overline{f_{\rm Edd}}_{\rm req} \geq f_{\rm quiet}$. Values $D_{\rm req} > 1$ identify an infeasible specified two-state history and are retained. An average growth requirement is not an instantaneous Eddington ratio or a uniquely measured duty cycle.

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency [1]:

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\rm ISCO}}}, \tag{9}$$

$$r_{\rm ISCO} = 3 + Z_2 - \mathrm{sgn}(a)\sqrt{(3-Z_1)(3+Z_1+2Z_2)}, \tag{10}$$

$$Z_1 = 1 + (1-a^2)^{1/3}\left[(1+a)^{1/3} + (1-a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{11}$$

The dimensionless spin is $a = cJ/(GM_{\rm BH}^2)$, where J is the signed angular momentum relative to the disk, and $r_{\rm ISCO}$ is in units of $GM_{\rm BH}/c^2$. The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Above unity, the maps adopt a phenomenological luminosity ansatz,

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \leq 1, \\ \epsilon_a f_{\rm Edd}\exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{12}$$

The logarithmic luminosity ansatz applies only on the super-Eddington branch. This is an illustrative parameter-scan prescription, not a relativistic slim-disk or spin-evolution calculation. If efficiency varies with time, the integrated growth depends on $\int[(1-\epsilon(t))/\epsilon(t)]f_{\rm Edd}(t)\,dt$; it cannot in general be represented by separately averaged efficiency and rate.

2 Catalogue construction

2.1 Scope and versioning

The catalogue uses scientific dataset versions rather than software releases. v1 is the original 23-object JADES broad-line catalogue. v2 expands the same mass-comparable analysis to 211 JWST broad-line objects. v3 retains all v2 objects and adds heterogeneous JWST-identified evidence classes. A source enters only one admission version; an internally heterogeneous source is assigned in full to v3. The fixed scope requires $z \geq 4$ and identification or candidate classification from JWST data. Objects merely followed up by JWST after a non-JWST discovery are outside the admission rule.

Source-family and measurement membership is frozen at the review cutoff of 3 September 2026; corrections to admitted measurements apply across affected versions, while new measurements require a declared subsequent version.

The v3 catalogue has 350 measurement rows, 338 provisional object records, and 337 provisional host records. Catalogue counts remain provisional. The manuscript inference samples exclude the affected records as specified below and in Section 5.4. It contains 247 broad-line AGN, 49 narrow-line candidates, 30 photometric candidates, ten high-ionization candidates, and two X-ray candidates. Preferred object-level evidence comprises 214 secure, 41 probable, 82 candidate, and one disputed classification. These labels describe the source papers' evidence and do not place all classes on a single purity scale.

2.2 Provenance, extraction, and identity

Each measurement carries stable measurement, physical-object, and host-system identifiers. Source-native values, adopted standardized values, uncertainty semantics, table locations, URLs, DOI or archive versions, archive SHA-256 hashes where available, extraction dates, and caveat tags are retained. The manual-extraction audit covers 26 hash-pinned extraction artifacts, while the provenance registry contains 38 records covering all 32 admitted source families. Independent checks cover all 244 numerical masses and both error fields. Auxiliary coordinate and contextual sources are attributed separately to the THRILS catalogue [9], the UHZ1 spectroscopy paper [20], and the versioned JADES DR3 GOODS-S prism catalogue.

Identity resolution preserves every admitted measurement while selecting exactly one preferred row per object. Its mass, redshift, evidence status, and method flags control the object-level comparison; alternate evidence remains recorded but is not averaged into that row. Preferences are explicit source-reviewed decisions, not a ranking by inferred growth pressure or a universal newest-publication rule. Previously adopted rows are retained unless a documented replacement is justified. For example, the CEERS-2782 group uses RUBIES-EGS-50052 because the source adopts its higher-signal-to-noise spectrum over the slit-loss measurement; NX10835 uses the newly available numerical NEXUS mass. UHZ1 uses the later full-dataset evidence assessment and remains outside numerical growth inference. The measurement–object link table records each decision and reason. Additional source-admission checks are summarized in Appendix A.1.

2.3 Mass comparability and analysis subsets

A measurement enters numerical growth analysis only when the source publishes a canonical numeric black-hole mass compatible with the declared comparison policy. It must have a numerical redshift, a named mass method, a numerical-mass comparability group, declared uncertainty semantics (including explicitly absent errors), an applied or otherwise resolved lensing treatment, and a resolved identity in the assembly registry. That last field denotes the recorded assembly decision, not closure of the stricter independent identity audit in Section 5.4. Masses inferred by assuming an Eddington ratio, indirect model ranges, and upper limits remain contextual observables. Published virial masses conditional on a broad component being a broad-line region are retained in the exploratory calculation but excluded from the primary comparison. This produces 244 eligible measurements and 237 eligible preferred objects. Of those objects, 236 use Balmer-line single-epoch virial masses and one uses a UV single-epoch virial mass. Their redshifts span 4.000–10.603 with median 5.110; standardized $\log_{10}(M_{\rm BH}/M_{\odot})$ spans 5.51–9.31 with median

7.18. The 101 objects without an eligible mass remain visible throughout the catalogue and atlas. Before the independent identity exclusions, the method-defined primary subset has 234 measurements and 227 preferred objects: an eligible preferred row must have secure or probable evidence, no conditional mass flag, and an affirmative primary-mass-comparison flag. In this release, all 227 method-eligible primary objects use Balmer-line single-epoch virial masses. Of the ten exploratory objects excluded from the primary sample, nine have candidate evidence (seven also have conditional broad-line interpretations), and GN-z11 has a UV estimator outside the primary method group. Missing reported errors do not remove an otherwise eligible point estimate; they prevent probability inference. These cuts define a method stratum, not identical calibration or uniform observational selection.

For manuscript inference we conservatively exclude every object record linked to any of the three open identity groups: six records linked to seven admitted measurements. Three excluded records have numerical masses (GDS_1210_9515, GS-8083, and GS-10013704); all three otherwise pass the primary method cuts. This yields 224 primary and 234 exploratory numerical objects. The other three excluded records already lack numerical masses. All source measurements and original catalogue flags remain available; exclusion does not assert that the records are duplicates or that their physical identities are resolved. The explicit exclusion registry and complete per-object selection are supplied with the manuscript analysis products.

Figure 1 distinguishes the 224 primary objects from the ten additional exploratory objects. The accounting panel also records 98 retained objects without masses and all six identity exclusions, without plotting excluded measurements. Every manuscript figure uses the publication selection; original full-catalogue figures remain available in the repository.

Table 1: v3 scale and analysis subsets.

Product or subset	Rows or objects
Measurements / provisional objects / hosts	350 / 338 / 337
Source-local observables	810
Growth-eligible measurements / objects	244 / 237
Method-eligible primary measurements / objects	234 / 227
Manuscript primary / exploratory objects	224 / 234
Identity-excluded object records	6
Catalogue-only object records	101
Alternate-measurement sensitivity pairs	7
Source-family caveat rows	32
Per-object parameter-map panels	676

3 Growth-model comparison

3.1 Reference model

We apply the relations in Section 1.1 to each eligible object. The reference ranking uses $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$, and $M_{\rm seed} = 10^2\,M_\odot$. Required average rates and seed masses follow Eqs. 6 and 7. We call the derived requirement *growth pressure*: it measures how demanding a specified history is, conditional on the published mass and model. A single merger factor and fixed efficiency do not resolve feedback cycles, changing accretion states, or stochastic mergers.

The spin-dependent $f_{\rm Edd}$–seed-mass maps and compatibility matrices use Eq. 12. The baseline rankings, duty-cycle diagnostics, and seed-redshift–mass maps retain $\epsilon = 0.1$; growth tracks retain their stated

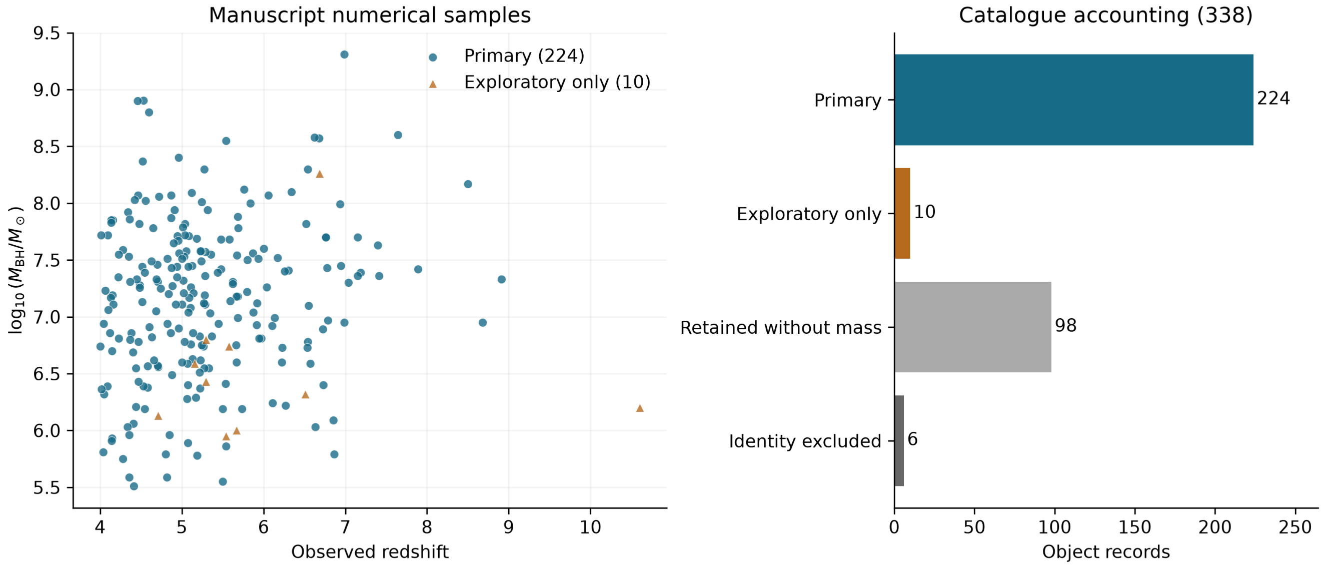

Figure 1: Manuscript selection. Left: 224 primary objects (circles) and ten exploratory-only objects (triangles). Right: mutually exclusive record counts, including 98 retained without masses and six identity exclusions. The exploratory numerical sample contains the primary sample and has 234 objects.

constant efficiencies. Thus compatibility above unity is conditional on a different efficiency model from the baseline required-$f_{\rm Edd}$ ranking. Figure 2 places the exploratory masses against reference tracks to show the mass and time scales. Primary and exploratory-only objects are distinguished; the tracks are reference histories, not fits to individual objects.

3.2 Uncertainty and ranking policy

Published asymmetric mass uncertainties in $\log_{10} M_{\rm BH}$ are propagated with 10,000 deterministic split-side draws per measurement and per preferred object. If a source gives no reported mass error, the atlas does not invent one. In particular, the twelve mass-bearing NEXUS objects have no published mass errors. Their repeated point values yield zero-width stored intervals; probabilities and uncertainty ranks are left unavailable, not interpreted as certainty. They are marked separately in uncertainty figures. The other 225 objects have reported mass errors in the full numerical catalogue; 222 remain in the manuscript exploratory sample after identity exclusion. Split-side sampling assigns half the probability to each side of the reported central value, using the respective error as the scale of a half-normal distribution; it approximates quoted intervals rather than reconstructing source posteriors. Redshifts are held fixed. Draws use seed 20260808 with a stable identifier-based random stream. For 10,000 draws the binomial Monte Carlo standard error of an estimated probability is at most 0.005 (approximately 0.0022 at probability 0.95); this numerical sampling error is separate from astrophysical uncertainty. Source-reported errors are retained as quoted and are not assumed to be purely statistical. For ZS7, the published 0.4-dex error already includes virial-calibration scatter. No additional scatter is added to that error. Separately reported method-scale systematics are stored and, where possible, mapped to sensitivity envelopes; the atlas does not assume that these terms are independent Gaussian errors.

The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

The primary scientific comparison is within object class and mass-comparability group. Global point and uncertainty ranks are retained only for navigation. The required-$f_{\rm Edd}$ comparisons in this manuscript use the physical growth requirement directly. The separate heuristic navigation score and tie policy are specified in Appendix B; they do not define the scientific conclusions. Compatibility is also calculated object by object across four seed families, three spin/efficiency cases, two merger cases, and three average accretion rates. For fixed $z_{\rm seed} = 30$, ϵ, $B_{\rm merge}$ and $f_{\rm Edd}$, a cell is compatible precisely when the required $\log_{10}(M_{\rm seed}/M_\odot)$ lies inside the inclusive seed interval: $[1, 2]$ (light), $[3, 4]$ (intermediate), $[4, 6]$ (heavy), or $[2, 6]$ (PBH-labelled). These are overlapping scenario ranges, not probability priors. A required seed below the interval indicates that this fixed history would overgrow the observed mass; it does not exclude the seed family under a shorter duty cycle. The PBH label specifies only a mass range at the common starting redshift, not primordial formation physics or a PBH abundance constraint. The source-level selection registry leaves inverse-probability weights unset because no common inclusion-probability model spans the 32 source families. Consequently, compatibility fractions are descriptive summaries of the admitted objects, not population frequencies.

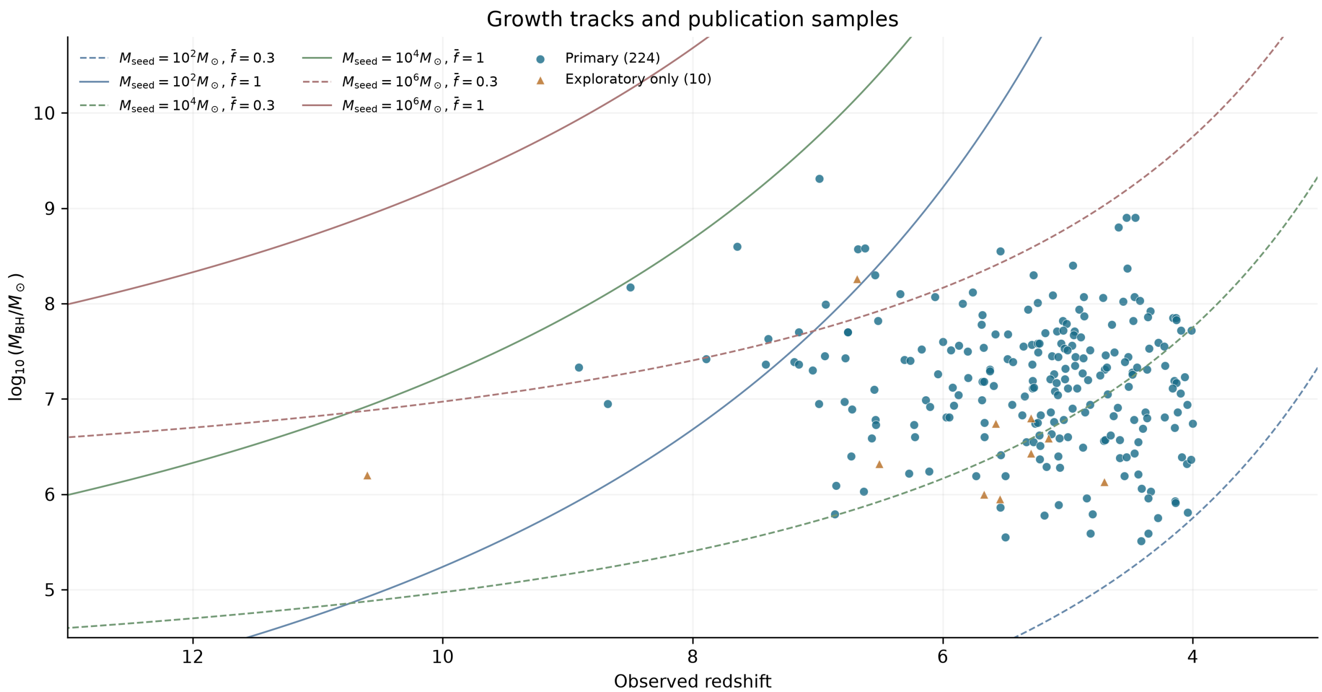

Figure 2: Reference growth tracks with the 224 primary and ten exploratory-only objects. Tracks use $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$, three seed masses and two average Eddington ratios. No excluded identities or mass-free records enter the plotted mass comparison; the reversed redshift axis spans $z = 13$ to $z = 3$.

4 Results

4.1 Growth pressure under the reference assumptions

In the 224-object manuscript primary sample, the required $f_{\rm Edd}$ distribution for a $10^2\,M_\odot$ seed has median 0.578 and interquartile range 0.487–0.682. Twelve objects require $f_{\rm Edd} > 1$ at their point estimates; none requires $f_{\rm Edd} > 2$. The manuscript exploratory 234-object sample has median 0.574, interquartile range 0.488–0.679, and 14 point requirements above unity. The high-pressure tail is not simply a mass ranking because less cosmic time can offset a lower observed mass. GN-z11, for example, has a lower standardized mass than many broad-line objects but has the second-largest required $f_{\rm Edd}$ at $z = 10.603$. Its UV virial estimator places it outside the primary comparison subset; this position is a conditional cross-method diagnostic, not a claim of equivalent mass-estimator reliability.

UNCOVER-20466 has the largest required average rate in both samples. Its point-estimate required

f_{Edd} is 1.452, and its Monte Carlo median is 1.453 with a 16th–84th percentile interval of 1.354–1.549. Table 2 reports the five largest primary-sample point requirements. GN-z11 is discussed separately as an exploratory cross-method comparison.

Table 2: Highest point-estimate required f_{Edd} for a $10^2\,M_\odot$ seed under the reference assumptions. Intervals propagate published mass errors only. All five objects belong to the 224-object manuscript primary subset; GN-z11 is excluded from this comparison.

Object	z	$\log_{10}(M_{\mathrm{BH}}/M_\odot)$	Required f_{Edd}	16th–84th percentile
UNCOVER-20466	8.500	8.17	1.452	1.354–1.549
GS-20057765	8.913	7.33	1.355	1.177–1.513
COSMOS3D-13852	7.646	8.60	1.314	1.247–1.384
RUBIES-EGS-55604	6.986	9.31	1.267	1.250–1.284
CEERS-1019	8.679	6.95	1.205	1.115–1.293

4.2 Robustness to reported mass uncertainty

Within the primary sample, eight objects remain above unity at the 16th percentile and six have $P(f_{\mathrm{Edd}} > 1) \geq 0.95$, compared with 12 above unity at their point estimates. In the exploratory sample, 14 objects have median required $f_{\mathrm{Edd}} > 1$, ten remain above unity at the 16th percentile, and 15 cross unity by the 84th percentile; eight have $P(f_{\mathrm{Edd}} > 1) \geq 0.95$. These probabilities are conditional on the source-reported errors and the reference growth assumptions; they do not add a common virial-calibration uncertainty or uncertainty in seed redshift, radiative efficiency, and merger history. These are descriptive counts within the stated subsets, not population fractions. Figure 3 displays the exploratory error distribution and separates missing-error cases from sampled intervals. Most objects lie below the reference threshold; the primary-sample counts above are calculated with the additional evidence and method cuts.

A separate full-catalogue publication-version check, performed on the original 227/237-object method-defined samples before the manuscript identity exclusions, replaces the frozen Baccus v1 measurements for 44 exact-ID, position-confirmed counterparts with the published Table 1 values. Five frozen objects lack a published-table counterpart. Both retaining and omitting those five objects preserve the original full-catalogue uncertainty counts and top five by required f_{Edd}. It is not a joint test of revised Baccus masses and the manuscript identity exclusions; the latter are assessed separately below. The 32 changed central masses shift by -0.10 to $+0.04$ dex. This is a measurement-version sensitivity test with fixed evidence/method taxonomy; it does not constitute a new admission of the complete published sample. The object-level comparison and three-scenario summary are distributed with the v3 science tables.

4.3 Model compatibility and measurement choice

Table 3 isolates changes to the reference history using the same preferred masses. Increasing the seed mass from 10^2 to $10^3\,M_\odot$ reduces the primary count above unity from 12 to four; a $10^5\,M_\odot$ seed removes all point requirements above unity. Delaying seed formation from $z = 30$ to $z = 20$ instead raises that count to 22. Lowering the efficiency to 0.05719 removes all point requirements above unity in both samples. These calculations change one assumption at a time; they are sensitivity benchmarks, not a joint distribution of physically realized histories. The seed endpoints bracket the light/heavy alternatives considered in analytic work [2]; $10^3\,M_\odot$ probes the transition between them. The $z = 20$ case tests a shorter available growth interval, and the lower efficiency is the ideal non-spinning thin-disk limit from Eq. 11. None of these choices is assigned a population weight or inferred from the data.

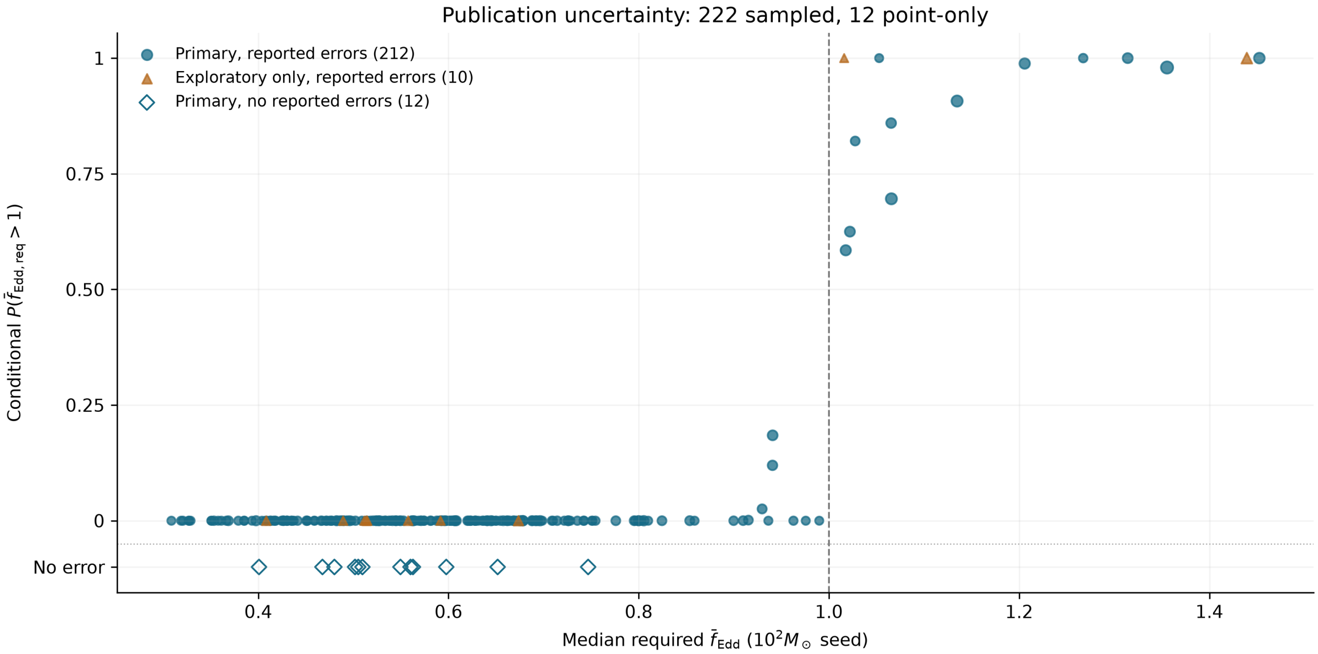

Figure 3: Publication-sample uncertainty. Circles show 212 primary objects with reported errors; triangles show ten exploratory-only objects with reported errors. Twelve primary objects without reported errors appear as open diamonds on the "No error" row, with no probability assigned. Point size for sampled objects encodes the 16th–84th percentile width.

Table 3: Point-estimate requirements above $\overline{f_{\rm Edd}} = 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use provisional object identities and do not include uncertainty tails.

Scenario	Primary (224)	Exploratory (234)
Reference	12	14
$M_{\rm seed} = 10^3 M_\odot$	4	5
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	22	24
$\epsilon = 1 - \sqrt{8/9}$	0	0

An explicit inclusion/exclusion sensitivity calculation compares these samples with the original 227/237-object sets. Excluding the six ambiguous records preserves the threshold counts and top-five ordering in every scenario in Table 3, as well as the reference 16th-percentile and $P \geq 0.95$ counts. The primary reference median changes from 0.574 to 0.578. Thus the reported early-growth tail does not depend on retaining the unresolved groups, although catalogue-wide uniqueness remains unsettled.

Figure 4 compares the scenario grid for the primary sample and the full 234-object exploratory sample. The latter includes the primary objects; the rows are not independent samples. Its dependence on spin/efficiency, merger boost, and seed interval illustrates why compatibility cannot be attributed to seed mass alone. These cells use the efficiency prescription of Eq. 12; they should not be read as extensions of the fixed-$\epsilon = 0.1$ reference ranking. Unavailable masses are excluded from numerical compatibility fractions.

Figure 5 shows seven alternate-measurement comparisons across six objects; CEERS-2782 contributes two alternatives. Alternate-minus- preferred mass shifts range from -0.600 to $+0.949$ dex, while the corresponding required-$f_{\rm Edd}$ shifts range from -0.083 to $+0.121$. All seven comparisons stay below unity at both point estimates. Thus these retained alternatives do not change reference threshold membership for the six tested objects, but they do not establish robustness for the untested high-pressure tail. The follow-up matrix distinguishes reference high-pressure objects, measurement-sensitive cases, and records still requiring a suitable mass.

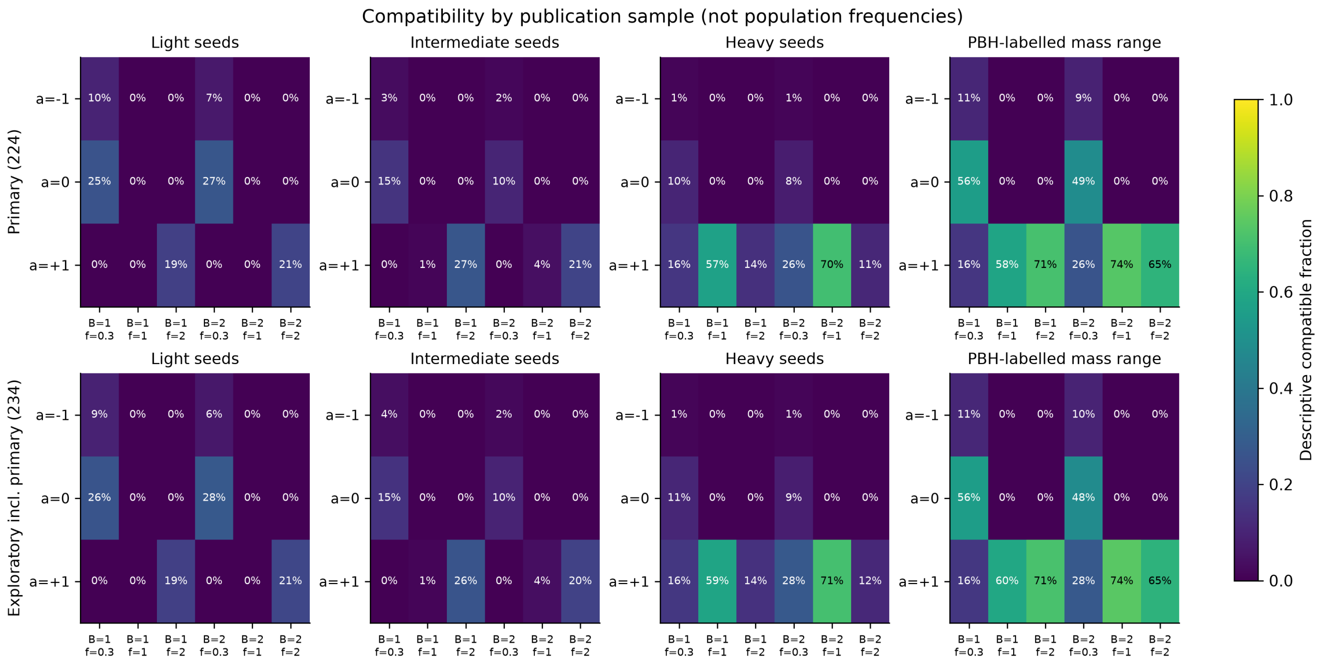

Figure 4: Compatibility in the 224-object primary sample (top) and the 234-object exploratory sample including it (bottom), after identity exclusions. Each cell is a descriptive compatible fraction within the labelled sample, not a completeness-corrected population frequency. Spin/efficiency cases use Eq. 12; seed intervals and merger boosts are fixed scenario definitions, not inferred formation channels.

5 Discussion

5.1 What the high-pressure tail establishes

The reference requirements above unity are concentrated at $z = 6.6214$–10.603. The principal growth constraint therefore comes from the early tail of the catalogue, while the lower-redshift majority supplies comparison objects.

The reference calculation identifies a small set of objects that are difficult to grow from $10^2\,M_\odot$ seeds at $f_{\rm Edd} \leq 1$ under the reference $\epsilon = 0.1$, $z_{\rm seed} = 30$ history. The result does not select a unique remedy. Earlier or heavier seeds, lower effective radiative efficiency, merger growth, or episodes above the reference accretion rate can each reduce the tension. The atlas is designed to expose these trade-offs for each object rather than collapse them into a single formation label.

This dependence can be quantified without changing the catalogue. At fixed seed mass, growth interval and merger factor, required $f_{\rm Edd}$ scales as $\epsilon/(1-\epsilon)$. The implemented ideal non-spinning thin-disk value $\epsilon = 1 - \sqrt{8/9} = 0.05719$ reduces every point requirement by a factor 0.54594 relative to $\epsilon = 0.1$: the largest becomes 0.793 and none exceeds unity. This sensitivity calculation does not infer actual spins or growth histories; it shows why the reference tail cannot establish a model-independent need for super-Eddington accretion.

The seed–accretion trade-off agrees qualitatively with Dayal's analytic comparison [2]. A numerical com-

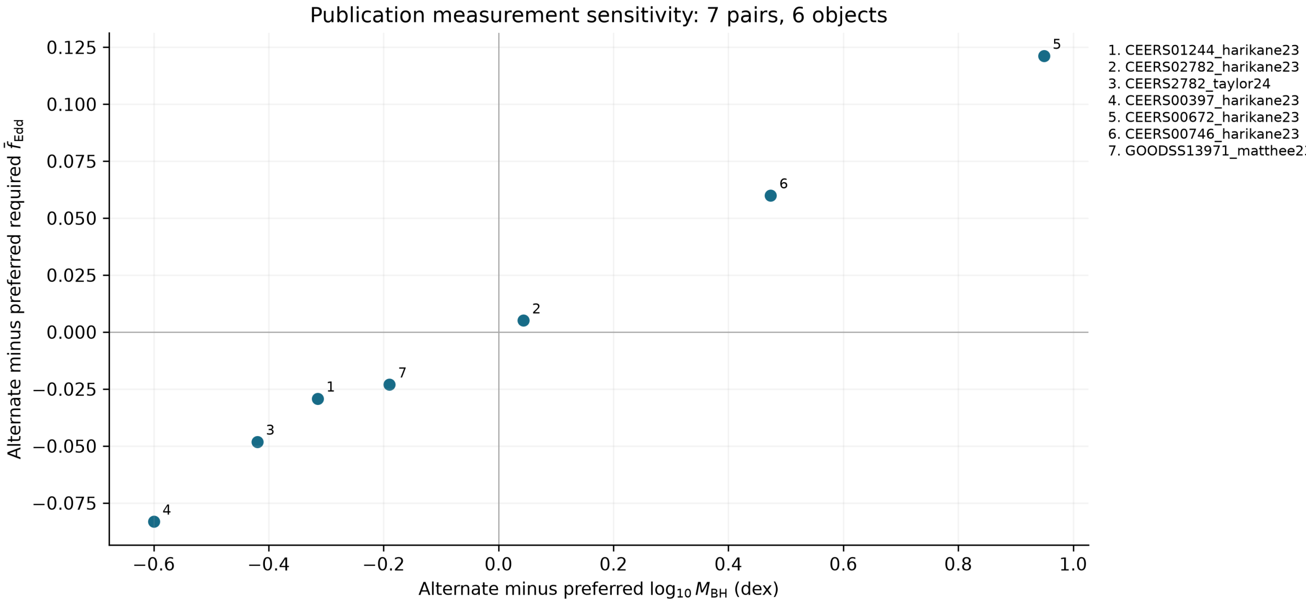

Figure 5: Seven alternate-measurement comparisons across six publication-primary objects. Numbered points are keyed to stable object and measurement identifiers.

parison of object counts would require identical membership, mass choices, and starting redshift; our reference uses $z_{\rm seed} = 30$ rather than 25. The added result here is the measured change in threshold membership and object ordering across explicit scenarios, with primary and exploratory evidence kept separate. This supports targeted observational tests rather than selection of a unique formation channel.

The most useful targets are therefore those whose placement is both demanding and stable to published mass uncertainty. Independent mass measurements, better separation of broad-line-region kinematics from scattering or outflows, and constraints on obscuration and lensing can change the physical interpretation more than another evaluation of the same analytic formula. The follow-up matrix makes that distinction explicit, but its heuristic ordering is not a calibrated calculation of expected information gain.

5.2 Contrasting object-level constraints

UNCOVER-20466 and GS-20057765 illustrate strong primary-sample constraints on the reference history: their required averages are 1.452 and 1.355, respectively, with reported-error 16th percentiles above unity. Increasing the seed to $10^3 M_\odot$ leaves their point requirements above unity (1.217 and 1.101), whereas $10^5 M_\odot$ seeds reduce them to 0.746 and 0.592. At the lower efficiency of 0.05719, the corresponding light-seed requirements are 0.793 and 0.740. Their masses constrain combinations of seed mass and sustained growth; neither object uniquely selects a seeding channel. For a quiet state with zero accretion and an Eddington-limited active state, these latter averages would require active fractions of about 79 and 74 per cent over the entire available interval. Falling below unity therefore need not imply an easily sustained fuel supply.

GN-z11 illustrates a different kind of constraint. Its high redshift yields a reference point requirement of 1.437 despite its smaller adopted mass. Delaying seeding to $z = 20$ raises that requirement to 1.880, exceeding UNCOVER-20466's 1.733 in that scenario. Thus even the identity of the most demanding object depends on the starting time. GN-z11's UV estimator excludes it from the primary sample; its probable evidence classification is permitted by the primary evidence cut. An independent, method-comparable mass would determine whether this apparent timing constraint survives the measurement assumptions. These examples distinguish sensitivity to the available growth time from sensitivity to mass methodology.

5.3 Value of the heterogeneous evidence layer

The 101 objects without eligible masses remain within project scope because the catalogue is a cross-paper evidence atlas as well as a growth calculator. Narrow-line, high-ionization, X-ray, and SED-selected candidates trace discovery channels that a mass-only broad-line catalogue would omit. They can be compared in redshift, evidence, survey, and follow-up need, but they are excluded from numerical growth claims until a suitable mass exists. This separation lets the atlas retain potentially important early accretors without converting qualitative evidence or model-dependent context into artificial precision.

5.4 Limitations

The dominant limitations are scientific rather than computational. The catalogue is a union of heterogeneous selections and cannot support pooled demographics without source-specific completeness models. Single-epoch virial masses share calibration and geometry uncertainties that are incompletely reported across papers. Two source-supported duplicate pairs have been reconciled without removing measurements. Three identity groups remain open: Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. The first requires target/spectral and preferred-mass reconciliation; the latter two require imaging and aperture checks. Unique-object and host counts remain provisional. All six affected records are excluded from manuscript inference; the inclusion/exclusion sensitivity above shows the effect on the reported results. This addresses their influence on the present analysis without claiming physical identity resolution. Future readmission requires source-backed spectral or imaging decisions, not wider matching thresholds. Finally, Eq. 4 compresses time-dependent accretion, feedback, mergers, and radiative-state changes into effective parameters. Its outputs should be read as conditional comparisons, not reconstructed histories. Reported-error probabilities condition on fixed growth parameters; marginalization requires justified parameter distributions and correlations, not just scenario maps. They are not seed-origin probabilities. The age relation neglects radiation; the supplementary $z_{\mathrm{seed}} = 3400$ plot is an extrapolation, not evidence for primordial formation or a physical radiation-era accretion history.

All admitted numerical mass/error fields have independent source checks; all 350 central redshifts and 323 available coordinate pairs are also checked against independent source evidence. Twenty-seven measurement rows lack coordinates. Redshift uncertainties, full source posteriors, and auxiliary observables are not exhaustively validated. Resolving the three groups is required before readmitting their records or claiming a final unique-object census. The literature cutoff is 3 September 2026; catalogue completeness refers to the declared source membership, not every JWST discovery. The A2744-QSO1 direct-mass measurement [38] awaits a new dataset version under the frozen-membership policy.

6 Reproducibility and data products

The repository provides pinned direct Python dependencies, five ordered notebooks, versioned catalogues, and hash manifests. Reproduction checks compare regenerated numerical tables and figures against committed baselines with explicit numerical and raster tolerances. Independent quadrature of the cosmic age integral and Runge–Kutta growth integration supplement these comparisons. The release includes per-object parameter maps, alternate-measurement checks, and source-linked follow-up information; implementation details and artifact inventories are documented in the repository.

Independent source checks cover all 244 numerical mass measurements and their error fields, all 350 central redshifts, and 323 available coordinate pairs. These establish source-value agreement within the declared checks, not the physical validity of every mass estimator or the uniqueness of every target. A separate manuscript exclusion check verifies that every open group is fully withheld from both inference samples and reproduces the stored selection and sensitivity tables. It runs in the public verification notebook. The stricter full-catalogue identity-resolution gate still reports the three groups in Section 5.4; the exclusion check does not replace it.

7 Conclusions

We have constructed a reproducible atlas with 338 provisional object records of JWST-identified accreting black-hole objects and candidates at $z \geq 4$. The main results are:

- 237 objects have a numerically eligible mass and 101 remain explicit no-inference records;
- after excluding the ambiguous identities, the primary 224-object sample has, under the light-seed reference history, 12 point requirements above $f_{\rm Edd} = 1$, eight above unity at the 16th percentile, and six with $P(f_{\rm Edd} > 1) \geq 0.95$; the exploratory counts are 14, ten, and eight;
- increasing the seed mass to $10^3 M_\odot$ reduces the primary point count to four, while delaying seeding to $z = 20$ increases it to 22; object ordering also depends on the assumed starting time;
- model compatibility is conditional on seed, efficiency, merger, and accretion assumptions, and the heterogeneous catalogue cannot support pooled demographic inference; lowering the fixed efficiency to the ideal non-spinning value removes all point-estimate requirements above unity; and
- complete provenance, identity history, uncertainty semantics, source caveats, follow-up priorities, and per-object visual products remain linked to the same frozen v3 dataset.

The manuscript results are unchanged in threshold membership and top-object ordering when the open identity groups are excluded. Final catalogue-wide object counts still require the identity work specified in Section 5.4. Within the present scope, the scientific priority is independent mass information for objects whose growth constraints are both demanding and sensitive to the assumptions, rather than a single global ranking of the heterogeneous catalogue.

A Source-family inventory

The input literature spans JWST broad-line samples [3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17, 18, 35, 36], host-selected broad-line candidates [10], an X-ray evidence history [19, 21], JADES narrow/high-ionization candidates [22], and GN-z11 high-ionization evidence [23]. Heterogeneous additions include high-ionization and narrow-line candidates [24, 25, 26, 30, 31], MIRI SED-selected candidates [27, 28], GHZ9 [29], compact blue broad-line emitters [32], additional UNCOVER candidates [33], MoM-BH*-1 [34], and GHZ4/GHZ7 [37].

A.1 Source-admission checks

The v3 additions include 15 NEXUS measurements from NIRCam/WFSS [35], 13 COSMOS-3D NIRCam-grism measurements [36], and GHZ4/GHZ7 [37]. These contribute 30 measurements and 29 object records. NX10835 lies 0.066 arcsec from an existing Mascia identifier at consistent redshift, so the new mass-bearing measurement becomes preferred for the same object.

The Scholtz et al. source audit provides another cardinality check. The paper reports 42 objects, whereas its source-native table contains 41 rows. Exactly 21 table rows satisfy $z \geq 4$; all are admitted. One is identity-linked to a broad-line object, yielding 20 distinct narrow-line candidates from that source.

JADES-NS-GS00099671 is retained with its published host and bolometric observables but without an invented black-hole mass.

B Catalogue navigation score

The point navigation score is

$$S = \min\left[100, 100\max\left(C\left(\frac{f_2 - 0.3}{1.2}\right), C\left(\frac{s_{0.3} - 4}{2.8}\right)\right) + 8C\left(\frac{z-6}{4}\right)\right], \tag{13}$$

where $C(x) = \min(1, \max(0, x))$, f_2 is the required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed, and $s_{0.3}$ is the required $\log_{10}(M_{\rm seed}/M_\odot)$ at $f_{\rm Edd} = 0.3$. This heuristic combines two growth diagnostics and a redshift term; it is not a probability. Ties use decreasing f_2, decreasing redshift, then stable identifier. The manuscript's required-$f_{\rm Edd}$ table is instead ordered directly by f_2.

C Release inventory

Canonical manifests record SHA-256 hashes for 85 v1, 461 v2, and 720 v3 artifacts. The current result inventory contains 1,236 artifacts. These inventories have different scopes: version manifests include dataset products as well as results. The repository records their individual entries and verification tolerances.

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